



Green Energy Production Using a Cheap Dual-Mechanism System



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Abstract

The purpose of the study is to develop a sustainable solution addressing the Grand Challenges of energy and environment by harnessing renewable resources. The chosen solution involved a hydroelectric generator that utilizes the natural flow of a river to rotate one-way turbines, coupled with Peltier coolers to generate additional power from the temperature gradient between river-cooled surfaces and solar-heated surfaces. A brief statement of what was done: We designed, constructed, and tested a prototype that integrates both hydro and thermoelectric systems. This prototype was developed to meet specific design requirements such as efficiency, simplicity, and environmental friendliness. One design requirement, in particular, that the solution excelled at was the energy production rate, which stated that the solution must generate at least 0.04 kW within 5 minutes. The chosen prototype produced 1.065 kW, all while ensuring continuous power generation without reliance on external fuels. A brief statement of major findings: the system produced electricity from both water flow and heat differential, validating the dual-source approach. Prototype Testing Results and Conclusions confirmed that the turbine system delivered solid performance under natural river conditions. While the thermoelectric generator (TEG) output was limited, it contributed additional power at no added operational cost, enhancing overall system efficiency. The major conclusions affirm that our solution can offer scalable, continuous energy solutions for off-grid applications, such as Toska Lake.

Introduction

Egypt is currently facing several urgent developmental issues, known as the Grand Challenges. Among these, three are especially critical: the need to improve the use of alternative energies, reduce pollution in air, water, and soil, and adapt to the effects of climate change. One of the primary contributors to these problems is Egypt's overwhelming reliance on fossil fuels for energy generation.

As illustrated in **Figure 1**, more than 90% of Egypt's electricity comes from fossil fuels, releasing vast quantities of greenhouse gases into the atmosphere and significantly worsening pollution levels and global warming (Our World in Data, 2024).

In response to these problems, Egypt and other nations have implemented large-scale renewable energy projects. One well-known national example is the Aswan High Dam, which generates hydroelectric power while also regulating Nile water flow.

Despite its advantages, including significant electricity output and agricultural support, the dam presents major drawbacks. These include ecosystem disruption, sediment buildup, and the displacement of local communities. Similarly, on a global scale, Morocco's Noor Solar Plant stands out as one of the world's largest concentrated solar power stations. It uses mirrors to focus sunlight and generate heat, which is then converted into electricity. While highly innovative, its limitations lie in its high land and construction costs, and its dependency on intense, consistent sunlight. Both examples informed us of what works and what challenges persist in existing renewable energy systems.

Before choosing our final solution, we established a clear set of design requirements. The system needed to produce at least 0.04 kilowatt-hours within five minutes of operation and be strong enough to withstand high water pressures without breaking. Environmental sustainability was also a core priority, so the materials and operations had to avoid causing harm or emitting pollutants. Finally, the model had to be compact, with a volume no greater than 0.5 cubic meters, to ensure it could be realistically constructed and tested.

With these design requirements in mind, our team developed a hybrid renewable energy system that combines a waterfall-based hydroelectric generator with a thermoelectric generator. The hydro component utilizes the kinetic energy of flowing water to spin a turbine, generating electricity rapidly and consistently. At the same time, the system leverages the temperature difference between the sun-heated surface and the cool water to power the thermoelectric generator. This dual approach not only increases energy output but also enhances reliability by ensuring power can still be produced even if one source fluctuates.

By integrating two renewable energy sources into a single compact and eco-friendly system, our solution directly addresses Egypt's Grand Challenges while offering a scalable alternative to traditional energy production.

To build the prototype that embodies our solution, we carefully selected materials that are not only functional but also environmentally responsible and locally accessible. The next section outlines the materials used in the construction of our prototype.

Materials & Methods

SMS40-4812 Fuji Electric's Stepper Motor	It is a 3.3V stepper motor. Used for energy production through turbines.	1	
TEC1-12706 Peltier Module	Used as a thermoelectric generator to take advantage of the heat differential as a result of solar energy, heat, and water cooling on hot and cold junctions.	2	
PVC Tubes (radii of 2cm and 1.25 cm)	Tubes that can be cut easily for a wide range of uses. That of r=4cm is used for turbine blades. That of r=1.25cm is used to direct water outside the Jerry can.	2	
Hollow threaded rod & its respective nuts	A rod of diameter 9mm, used for holding the turbine and the ball bearings in their position together.	1	
Jerry can	A container for the water and the turbine is to be placed.	1	
DC-DC Buck converter LM2596	Used for the final phase in AC-DC conversion by smoothing out DC as well as stepping down the voltage for practical use in charging the batteries included.	1	
18650 3.7V 1400mAh Lithium-Ion battery	A rechargeable battery & a good choice in terms of sustainability & safety.	1	

Table 1: Materials used in construction of the prototype

Phase 1: Building the Cross-flow Turbine (See Figure 2)

1. Identify cross-flow turbine components: thin circular sheets and blades made from halved 4 cm PVC pipes.
2. Cut each half-tube into 10 cm long sections (total of 8 blades), sanding for smoothness.
3. Create disks from CD drives + 7.5 cm radius recycled plastic circles.
4. Stick blades between two disks (plastic outside, CD inside) using Epobond A&B glue.
5. Ensure blade inlet angles are exactly 20° for optimal flow efficiency.
6. Finalize by assembling all parts together to complete the turbine body, as illustrated in Figure.

Phase 2: Preparing the Jerry-Can & Fixing the Turbine

1. Determine optimal turbine position, center of jerry-can (intersection of diagonals).
2. Cut a large hole on the side opposite the jerry-can opening (the flow-out side).
3. Insert two ball bearings in the hole and fix them to support a metal hollow rod.
4. Slide the turbine onto the rod with nuts securing it on both sides.
5. Make sure one end of the rod extends enough to hold a large gear for motor contact.

Phase 3: Setting Up the Stepper Motor & Its Gears

1. Glue a 46-teeth gear to the turbine's metal rod.
2. Attach a 10-teeth gear to the motor's shaft.
3. Mount the motor on a small elevated stage for gear alignment.
4. Ensure gears mesh to convert 1 turbine rotation into 4.6 motor shaft rotations.

Phase 4: Setting Up the Peltier Modules

1. Use a PVC pipe for water outflow; cut vertically to create a half-pipe section.
2. Attach stainless steel metal over the cut area to host the cold junction of the Peltier.
3. Apply thermal paste between the metal and Peltier modules for heat transfer.
4. Add another thermal-pasted sheet on top for the hot junction.
5. Use convex lenses to concentrate sunlight onto the hot junction at optimal distance.

Phase 5: Setting Up the Electrical Circuits

1. Use a full-bridge rectifier with Schottky diodes to convert AC to pulsating DC.
2. Add 1000µF 16V capacitors to smoothen the DC current.
3. Combine power from both turbine and Peltier modules.
4. Feed the final output into a DC-DC buck converter for voltage regulation.

Design Requirements

1. **Fast in Producing Energy** – Must generate at least 0.04 kW within 5 minutes.
2. **High Strength** – Able to withstand harsh conditions without breaking.
3. **Energy Storage** – Must store the generated energy in a battery.
4. **Environment-Friendly** – Should not cause significant harm to the environment.
5. **Volume** – Total volume should not exceed 0.5 m³ for portability.

Test plan

1. **Safety precautions:** Make sure the entire team is working safely through the use of coats, gloves and masks to avoid the toxic emissions.
2. **Energy Output:** Place in water flow, start timer, connect to multimeter, log energy every 30 sec for 5 mins. Repeat 3x. Must average ≥ 0.04 kWh.
3. **Durability:** Drop from 1m onto soft surface, spray with water (10 mins), expose to sun (2 hrs), re-run the energy output test to confirm no damage or performance drop.
4. **Energy Storage:** Connect to a rechargeable battery, run 10 mins, measure stored energy, use it to power a small device such as a fan or LED.
5. **Eco-Friendliness:** Check materials for recyclability/non-toxicity, test noise, emissions, and heat, must be clean, quiet, and safe.
6. **Volume Check:** Measure volume of multiple instance cuboids, confirm total volume ≤ 0.5 m³.

Negative Results

Initial testing of the hydroelectric turbine system revealed critical limitations in power generation. The prototype struggled to maintain a consistent voltage, dropping to as low as 5 volts, as shown in **Figure 3**. This low voltage resulted in reduced power output and demonstrated inefficiencies in the system's energy conversion. The system's overall efficiency was calculated at just **7.8%** (see **Table 2**), which is primarily due to mechanical friction, losses within motor, and unstable water pressure. These challenges underscored need for enhanced flow control & turbine optimizing.

Positive Results

Subsequent modifications to the prototype resulted in significant performance improvements. One key enhancement was the integration of a hose reducer, which narrowed the water flow path and increased flow velocity from 2.83 m/s to 11.33 m/s, in accordance with Bernoulli's Principle. This change allowed turbine to convert more hydraulic energy into kinetic & electrical energy. These mechanical adjustments led to a substantial rise in electrical output. Voltage levels became more stable and consistently higher, peaking at 17 volts for a short time, and in normal conditions, averaging above 10 volts, as shown in **Figure 4**. From here, power was calculated using Ohm's Law that states $P=V^2/R$, and the power was reached up to 25.13 watts at the 17V peak. This all reflects the system's improved energy generation efficiency across the test duration, shown in the results in **Table 3**. Additionally, the turbine produced a total of 538 Joules over 65 seconds. The average flow rate remained at 0.757 liters/second, ensuring consistent operation. These results confirm the system's validity and its capability to be a low-cost, renewable energy solution with further refinement.

As for the Peltier modules, they have proven their efficacy in generating electricity by a heat differential of 40 degrees Celsius, by producing a DC potential difference of 1.3V, along with an average of 480mA. The power output is calculated to be at an average of 0.6 Watts.

All the measurements for the turbine were measured at an error percentage value of ±1.0%, and an error percentage value for the Peltier modules' readings of ±0.5% for the voltage and ±1.8% for the amperage, according to the datasheets containing information about the multimeter used.

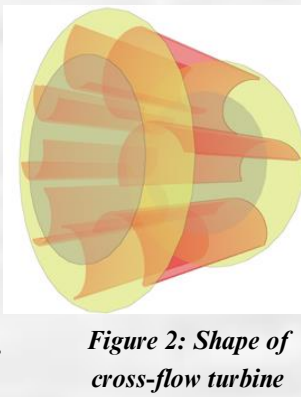


Figure 2: Shape of cross-flow turbine

Analysis

Electric power is the rate at which electricity does work, at a point in time, which is measured in watts; corresponding to Joules/second. (**ES.1.09**)

The average value of resistance, of 11.5 Ohms, was safe to be considered, knowing that it doesn't fluctuate as much. It is used in order to calculate the power without having to measure the voltage and the amperage simultaneously. Depending on Ohm's law, an approximate value of the power can be deduced from the voltage alone, in a series of equations, as follows:

$$\frac{V}{R} = I, \quad I = \frac{V}{R}, \quad VI = P, \quad P = \left(\frac{V}{R}\right)(V) = \frac{V^2}{R}$$

It's known that displacement is calculated in a velocity-time graph, where $V = \frac{d}{t}$, by calculating the area under the curve. The same method can be used to calculate the number of joules in power-time graphs, as $P = \frac{J}{t}$. (**ME.1.02**)

In order to determine the number of joules produced throughout the minute and 5 seconds, the area of each trapezium under the curve can be calculated as shown in **Figure 6**. The equation for the area is given as $A = \left(\frac{a+b}{2}\right)(h)$.

, knowing that a and b are 2 vertical magnitudes of power for each two adjacent points on the curve. By summing the magnitudes of areas of all the trapezia present, the number of joules produced was estimated at 538 joules. For the energy production by Peltier modules, the refractive property of paraboloid-like lens was taken advantage of, by analyzing each surface of lenses into 2 parabolas, with foci & other properties determined. (**MA.1.07**).

It was necessary to examine the standard values of specific heat capacity of different materials in order to determine which material conducts heat the best. A low specific heat capacity meant a much, much better heat conduction, which is necessary for heating the hot junction and cooling the cold junctions. Using copper sheets was the best option as a conductor of electricity, with its specific heat capacity of **0.385 J/g °C**. The amount of heat transferred was estimated for the efficacy of the Peltier modules, using the heat capacity equation, which states: $Q = mc\Delta T$. (**PH.1.10**)

It is extremely important to note that stepper motors generate alternating current which is yet to be converted to DC through rectification and stabilization. The ratio between the apparent power, which is the resultant in all the graphs and through the Ohm-law method, is actually apparent value, while real value is always beneath that by a certain number called the power factor. The power factor is a dynamic variable. It can be corrected by using capacitors with certain capacitance according to the initial power factor, the frequency of the AC wave, and other properties of electric current.

In order to convert a larger portion of the energies involved in the flow of the water hose into kinetic energy, it was necessary to decrease the cross section of the hose. That's the mathematical representation of **Bernoulli's principle**, known as **Bernoulli's equation**: $\rho gh + \frac{1}{2} \rho v^2 + P = \text{constant}$ (**PH.1.09**)

The internal pressure P is constant, meaning that the main factors in play here are the hydrostatic pressure (ρgh) and the kinetic energy or the kinetic pressure per unit volume, ($\frac{1}{2} \rho v^2$). (**PH.1.08**) After reducing the diameter of the hose from **2cm to 0.9cm** with a reducer, the circular cross-sectional area has decreased to less than the quarter of its original value, and the flow velocity can be calculated, knowing the volume flow rate, which is equal to **0.757 liters/sec**, or **757 cm³/sec**. Here, the equation of continuity can be utilized. (**PH.1.09**)

The overall efficiency of the turbine system can be calculated, by dividing the output (electrical power) over the input (hydraulic power), whose equation is $\eta_{\text{overall}} = \frac{P_{\text{electrical}}}{P_{\text{hydraulic}}} = \frac{\text{Volts} \cdot \text{Amperes}}{\rho \cdot g \cdot h \cdot Q}$. It is important to note that the h in the equation is the **theoretical head**, which corresponds to the height which theoretically accelerates the flow velocity to be at the final value of 11.89 m/sec. It does not correspond to the actual vertical distance between the jet and the turbine. Nevertheless, it can be calculated using the known kinematic equations, with the assumption that the initial velocity is zero, and $g = 9.81 \text{ m/sec}^2$

$$h = \frac{(v_{\text{final}})^2}{2g} = \frac{(11.89 \text{ m/sec})^2}{2 \cdot 9.81 \text{ m/sec}^2} = \frac{141.37 \text{ m}^2/\text{sec}^2}{19.62 \text{ m/sec}^2} = 7.20 \text{ meters}$$

Having obtained value of the **head**, it is possible to make an estimate of overall efficiency of turbine, as follows:

$$\eta_{\text{overall}} = \frac{P_{\text{electrical}}}{P_{\text{hydraulic}}} = \frac{\left(\frac{(V_{\text{MAX}})^2}{R}\right)}{\rho \cdot g \cdot h \cdot Q} = \frac{8.52 \text{ Watts}}{1000 \text{ kg/m}^3 \cdot 9.81 \text{ m/sec}^2 \cdot 7.20 \text{ m} \cdot 0.000757 \text{ m}^3/\text{sec}} = 15.9\%$$

The efficiency value of kinetic energy transmitting from water jet to the turbine is known as the hydraulic efficiency, and it depends heavily on the geometrical structure of the turbines in allowing the water to exert the largest possible fraction of its force over the turbine blades.

The angular velocity of the turbine was estimated to be 61.261 radians/second, which, by multiplying by a radius of 7.5cm, we obtain a linear velocity of 4.59 m/sec.

The equation for the theoretical work done by the turbine shaft is as follows, and it is called Euler's turbine equation. Assuming that all friction and all imperfections are not present, the equation states:

$$W_{\text{shaft}} = \dot{m} \cdot U \cdot (C_{x1} - C_{x2}) \quad (\text{Dendy Adanta et al, 2020})$$

The geometry of blades consists of slanted half-circles, where water enters tangentially & exits at an opposite direction. The blades redirect flow entirely. The relative velocity at inlet is equal to the negative of the relative velocity at outlet, and their vector values are inverse to each other because of the 180 degrees flip. Keep in mind that the symbols with subtext 1 are properties of inlet, and those of subtext 2 are properties of outlet. C indicates absolute velocity, U indicates blade velocity and W indicates velocity relative to moving blades.

$$V_{r1} = -V_{r2}, \quad V_{r2} \cos(\beta_1) = -V_{r1} \cos(\beta_1)$$

Therefore, we can calculate C_{x1} and C_{x2} as follows:

$$C_{x1} = V_{r1} \cos \beta_1 + U, \quad C_{x2} = -V_{r2} \cos \beta_1 + U = V_{r2x} + U$$

$$|V_{r1}| = |V_{r2}| = V_r$$

The values were plugged into the equation, and the horizontal part of the absolute outlet velocity was compensated for that of the inlet velocity for clarity. The mass flow rate equals 0.757 kg/sec, the blade linear velocity is 4.59 m/sec, the flow velocity C_1 equals 11.89 m/sec and after using **Figure 7**, V_r was 7.7377 m/sec, and finally β_1 was equal to 31.704 degrees.

$$W_{\text{shaft}} = \dot{m} \cdot U \cdot (V_r \cos \beta_1 + U) - (-V_r \cos \beta_1 + U) = \dot{m} \cdot U \cdot (2V_r \cos \beta_1)$$

$$W_{\text{shaft}} = \left(0.757 \frac{\text{kg}}{\text{sec}}\right) \left(4.59 \frac{\text{m}}{\text{sec}}\right) \left(2 \cdot 7.7377 \frac{\text{m}}{\text{sec}} \cos 31.704^\circ\right) = 45.74 \text{ Watts}$$

Knowing that the hydraulic efficiency equals the shaft power produced by the power exerted by the hydraulic power of the water jet, it equals 85.5%, as clarified in the final equation here:

$$\eta_{\text{hydraulic}} = \frac{W_{\text{turbine shaft}}}{P} = \frac{45.74 \text{ Watts}}{\rho \cdot g \cdot h \cdot Q} = \frac{45.74 \text{ Watts}}{1000 \text{ kg/m}^3 \cdot 9.81 \text{ m/sec}^2 \cdot 7.20 \text{ m} \cdot 0.000757 \text{ m}^3/\text{sec}} = 85.5\%$$

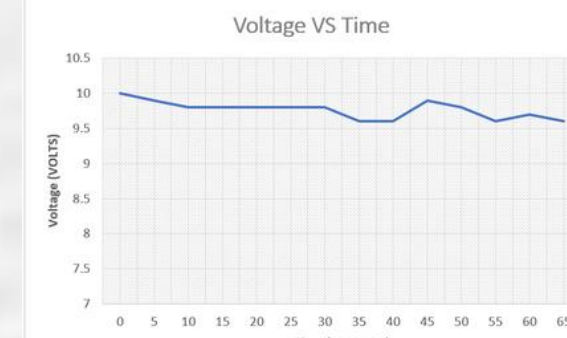


Figure 5: Voltage recorded in 2nd testing after improvement

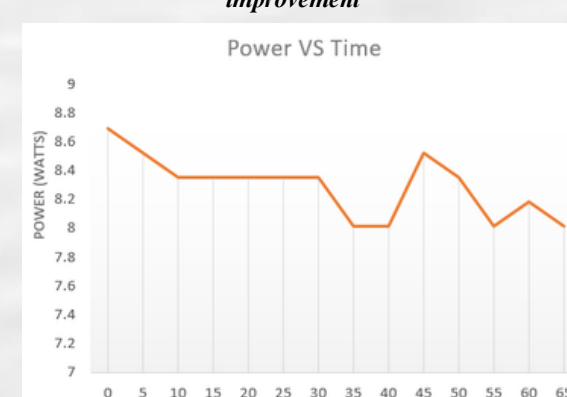


Figure 6: Power calculated according to the resistance & voltage

Conclusion

The problem to be solved centered on Egypt's overreliance on fossil fuels and the urgent need for sustainable, eco-friendly alternatives to electricity production. Our prototype testing results, derived from a hybrid system combining a cross-flow hydro turbine with thermoelectric generators (TEGs), demonstrated that river flow can reliably power the turbine, while cooling the (TEGs), improving the heat differential, producing usable electricity. From our analysis, the system proved to be functional, efficient, low-cost, and easy to replicate. The combined hydro and thermoelectric setup offers consistent performance and self-sufficiency, and from here we can consider it a valid solution that will succeed on a real-life scale in areas that offer both water flow and sunlight, such as the Toshka Lakes region. This solution addresses all design requirements: it's efficient, eco-friendly, cost-effective, made from available materials, stores energy, and is user-friendly and portable, as well as the fact that the results showed the validity of the graphs shown in the analysis, as the prototype was indeed able to produce 538 joules in 65 seconds. Compared to prior solutions like the High Dam or Noor Plant, our system is more scalable, affordable, and adaptable to isolated communities without large infrastructure, all while theoretically providing a higher energy output.

To further improve the results, some issues must be addressed in the recommendation section.

Recommendations

The experience gained throughout the construction of the prototype has necessitated the inclusion of specific recommendations for future improvement and sustainability. There have been many pitfalls during the earliest stages during the prototype building, and the following recommendations illustrate the correct steps and protocols needed in avoiding them. The recommendations are:

- Oscilloscopes are expensive and hard to come by in Egypt, but they are needed to calculate the frequency and the power factor of the AC output, in order to determine the real power consumed from the current under the given load. This will help with correcting the power factor through choosing the correct capacitance of the capacitors after the full-bridge rectification using the diodes.
- Using microcontrollers and microchips such as Arduino-based chips, to take measurements and send real-time feedback of the water jet's flow rate and velocity, the turbine's RPM and the temperatures at the hot and cold junctions of the Peltier modules.
- The Toshka Lakes region presents an ideal environment for future deployment of a large-scale version of the prototype, as it offers both strong solar irradiance and a continuous freshwater source, making it perfect for activating both the hydroelectric and thermoelectric components of the system, additionally, choosing this location is strategic due to its remoteness and agricultural importance, making it a strong candidate for decentralized energy support in non-centralized areas of Egypt such as upper Egypt.
- For future large-scale applications, the system should be tested using more durable turbine materials and larger-scale water flow setups to reflect real environmental conditions. Since high-efficiency thermoelectric modules and industrial-grade components were too costly for our prototype, they should be introduced gradually in the real-life application to boost performance. The design should also be adapted into a modular setup that allows for easier assembly and maintenance in rural areas. This would make the system more practical for real deployment without needing massive infrastructure.

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Table 3: Illustrating Voltage, Power & Time simultaneously for 65 seconds